Karim Hassan

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EDUCATION

MEMORIAL UNIVERSITY
BE. IN COMPUTER ENGINEERING
St. John's, NL

LINKS

Website:» Karim-Hassan
Github:» Karim-Hassan
LinkedIn:» Karim-Hassan

SKILLS

PROGRAMMING

Over 10000 lines:
C++ • Python • Java • Assembly
MATLAB • SQL

Over 5000 lines:
Javascript (Node.js) • Visual Basic
HTML • CSS • C

Familiar:
PHP • React • LaTeX

SOFTWARE TOOLS

MySQL• GitHub • Jira • VMware
Git • Azure

Familiar:
Bitbucket • MongoDB

AWS

S3, Lambda, API Gateway

OPERATING SYSTEMS

Microsoft Windows • Linux
Familiar:
Unix • Mac-OS

SOFT SKILLS:

Leadership, Self-Motivated
Team-player, Detail-oriented
Fast-Learner, work under pressure

EXTRA CURRICULAR

IEEEXTREME

Participated in the IEEE 24Hr coding challenge and finished within the top 40 teams nationally

INTERESTS

Game Development • AI •
Ethical-Hacking • Computer Vision
Cyber Security

PROFESSIONAL EXPERIENCE

CRASHBAY | Data engineer

May 2025 - Present | Ottawa, ON

- Engineered a centralized Azure SQL data warehouse by designing API-driven ETL data mapping from multiple operating systems, powering real-time visualization in Power BI.
- Automated end-to-end data extraction pipelines to continuously feed structured claims and financial data to external AI vendor APIs, eliminating manual reporting bottlenecks.
- Designed and implemented new SQL schemas, including UAT environments, to support data integration, testing, and analytics workflows across systems.
- Developed data transformations to sanitize multi-regional financial records, normalizing and resolving cross-table structural anomalies to ensure high-fidelity reporting.
- Migrated and enriched historical records from Google Sheets into Azure data warehouse, appending geospatial attributes (lat, long, Google Place ID) to support dashboards and operations reporting.

J.D. IRVING | Programmer Analyst

Dec 2021 - Apr 2022 | Moncton, NB

- Analyzed paper mills data to enhance the efficiency and reduce the downtime by 10%
- Design, implement, and manage the organization's data strategy, as well as develop software and analytic solutions to improve the use of data across the organization.
- Developed Data as a Service (DaaS) platform with Node.js, TypeScript, and REST APIs to provide streamlined access to the organization's data.
- Leveraged Data as a Service to create data-driven web applications and dashboards with Node.js, TypeScript, MongoDB, and Material-UI.

STEMWORLD | Software Developer | Team Lead

May 2021 - Aug 2021 | Ottawa, ON

- Build end-to-end Robotic kits on ATmega microcontrollers using C & assembly.
- Developed a web registration system & structured a MySQL database and main connection object for it.
- Worked in the machine learning department. Planned, designed, and developed a machine vision model for a mobile adaptation of Google's quick draw game using Node.js, Bootstrap framework, REST API, JS, HTML, and CSS.
- Used (PyCharm) Python's framework 'TensorFlow' to train and predict. Moreover, NumPy Matplotlib, and pandas for multi-dimensional arrays, data visualization, and data analysis were used.

MARINE INSTITUTE | Data Analyst

Sep 2020 - Dec 2020 | St. John's, NL

- Developed a code that combine and analyze 3 different datasets to track a pattern in the passengers' behavior during evacuation emergencies on ships.
- Extracted distances between IR beacons using AutoCAD to create a new database.

PROJECTS

EASTERN EDGE ROBOTICS | Junior Developer

Sub-marine Robot Project Sep 2019 – Aug 2020 | St. John's, NL

- Developed embedded system communication on Arduino using Python & C++
- Implemented the control GUI for the ROV written in Python and C++

PARADIGM HYPERLOOP | Junior Developer

Sep 2019 – Aug 2020 | St. John's, NL

- Architected object-oriented codebase for the vehicle's embedded systems
- Wrote drivers for custom PCBs using SPI, I2C, CANBus and UDP
- Developed a Python tool to automate uploading and organizing testing data on Cloud
- Utilized SciPy to calculate aerodynamic properties in real-time to reduce its runtime